Engineering Mathematics: Eigenvalues and Eigenvectors

1. Fundamental Definitions

To analyze the transformation characteristics of a square matrix, we use the concepts of characteristic matrices and equations.

- **Characteristic Matrix**: For a square matrix A of order n, the matrix $(A - \lambda I)$ is known as the characteristic matrix, where λ is a scalar variable and I is the identity matrix.
- **Characteristic Equation**: The equation obtained by setting the determinant of the characteristic matrix to zero, $|A - \lambda I| = 0$, is the characteristic equation.
- **Eigenvalues**: The roots of the characteristic equation are called Eigenvalues (or Latent/Characteristic roots).
- **Eigenvectors**: A non-zero vector X that satisfies the homogeneous linear system $(A - \lambda I)X = 0$ is the Eigenvector corresponding to the eigenvalue λ.

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2. The Working Algorithm

The logical flow for solving these problems consists of three primary stages:

1. **Formulate the Characteristic Equation**: Construct the determinant $|A - \lambda I|$ and set it to zero.
2. **Determine Eigenvalues**: Solve the resulting polynomial for λ.
3. **Solve for Eigenvectors**: For every individual λ value, solve the system $(A - \lambda I)X = 0$ using Gaussian elimination or row transformations.

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3. Step-by-Step Transformation Example

We will find the eigenvalues and eigenvectors for matrix A:

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

Step 1: Solve the Characteristic Equation

Set $|A - \lambda I| = 0$:

$$\begin{vmatrix} 5 - \lambda & 4 \\ 1 & 2 - \lambda \end{vmatrix} = 0$$

Expanding the determinant:

$$(5 - \lambda)(2 - \lambda) - (4 \times 1) = 0$$

$$10 - 7\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 7\lambda + 6 = 0$$

Factoring the quadratic:

$$(\lambda - 1)(\lambda - 6) = 0$$

The **Eigenvalues** are $\lambda_1 = 1$ and $\lambda_2 = 6$.

Step 2: Find Eigenvector for $\lambda = 1$

Substitute $\lambda = 1$ into $(A - \lambda I)X = 0$:

$$\begin{bmatrix} 5-1 & 4 \\ 1 & 2-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Applying row transformation $R_1 \longrightarrow R_1 \times (\frac{1}{4})$:

$$\begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} (R_1 \to \frac{1}{4} R_1) \longrightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Applying $R_2 \longrightarrow R_2 - R_1$:

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} (R_2 \to R_2 - R_1) \longrightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

This gives the equation $x_1 + x_2 = 0$, or $x_1 = -x_2$.
Let $x_2 = -k$, then $x_1 = k$. For $k = 1$:

$$X_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Step 3: Find Eigenvector for $\lambda = 6$

Substitute $\lambda = 6$ into $(A - \lambda I)X = 0$:

$$\begin{bmatrix} 5-6 & 4 \\ 1 & 2-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Applying $R_2 \longrightarrow R_2 + R_1$:

$$\begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} (R_2 \to R_2 + R_1) \longrightarrow \begin{bmatrix} -1 & 4 \\ 0 & 0 \end{bmatrix}$$

This gives the equation $-x_1 + 4x_2 = 0$, or $x_1 = 4x_2$.
Let $x_2 = 1$, then $x_1 = 4$:

$$X_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$